Notes for Chapter 1

Matrices and Systems of Equations

1.1:Systems of Linear Equations

Elementary Row Operations

I. Interchange two rows.
II. Multiply a row by a nonzero real number.
III. Replace a row by its sum with a multiple of another row.

1.2: Row Echelon Form

Row echelon form (Gaussian elimination):

A matrix is said to be in **row echelon form**

(i) If the first nonzero entry in each nonzero row is 1.
(ii) If row k does not consist entirely of zeros, the number of leading zero entries in row $k+1$ is greater than the number of leading zero entries in row k.
(iii) If there are rows whose entries are all zero, they are below the rows having nonzero entries.

Example:

The following matrices are in row echelon form:

$$\begin{bmatrix} 1 & 4 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 3 & 1 & 0 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The following matrices are not in row echelon form:

$$\begin{bmatrix} 2 & 4 & 6 \\ 0 & 3 & 5 \\ 0 & 0 & 4 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Reduced row echelon form (Gauss-Jordan reduction):

A matrix is said to be in **reduced row echelon form** if

(i) The matrix is in row echelon form.
(ii) The first nonzero entry in each row is the only nonzero entry in its column.

Example:

The following matrices are in reduced row echelon form:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 2 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Homogeneous Systems (Ax=0):

Any homogeneous system is **consistent**. Because there is a special solution **x=0.**

Comparison between Ax=b and Ax=0:

Ax=b:

For a linear system of m equations and n unknowns, the number of solutions can be summarized as follows.

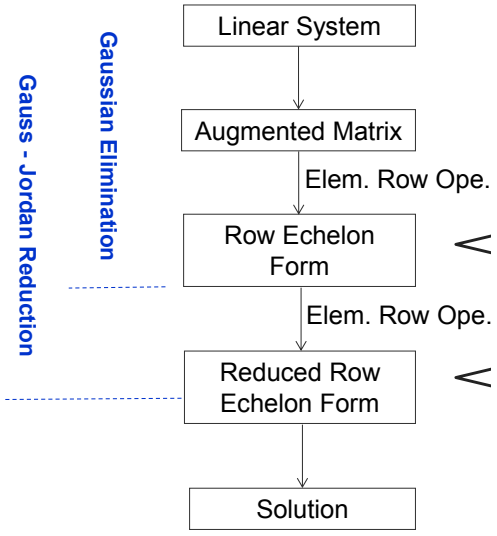
	$m > n$	$m = n$	$m < n$
no solution	most likely	little chance	little chance
unique solution	little chance	most likely	impossible
infinity many solutions	little chance	little chance	most likely

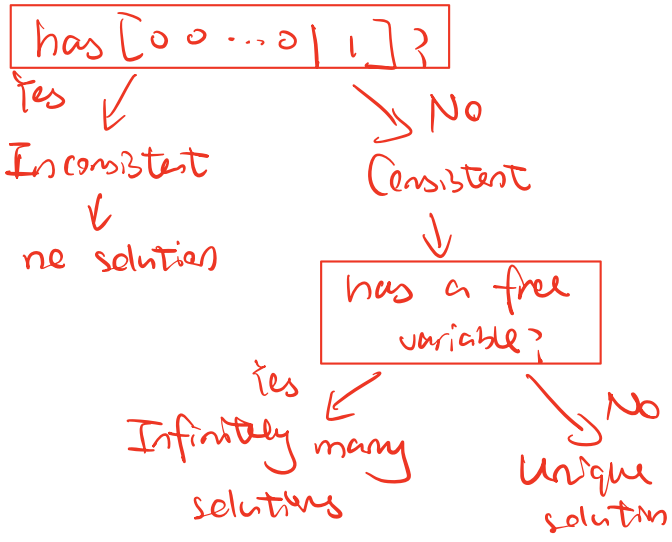
Ax=0:

For a homogeneous system of m equations and n unknowns,

	$m > n$	$m = n$	$m < n$
no solution	impossible	impossible	impossible
unique solution $(0, 0, \cdots, 0)$	most likely	most likely	impossible
infinity many solutions	little chance	little chance	for certain

How to solve a linear system, how many solutions it has?





Example: Solve the following linear system

$$\begin{aligned} x_1 + x_2 + x_3 + \ x_4 + \ x_5 &= 2 \\ x_1 + x_2 + x_3 + 2x_4 + 2x_5 &= 3 \\ x_1 + x_2 + x_3 + 2x_4 + 3x_5 &= 2 \end{aligned}$$

Other important points:

1. Trivial solution means x=0. Nontrivial solution means the solution the does not equal to 0. (**Trivial solution: 零解； Nontrivial solution: 非零解**)
2. When m<n and the linear system has infinitely many solutions, you should be able to express the solution set by using free variables, for instance, $(5, 3 + 2\alpha, 6\beta, \alpha, \beta), \alpha, \beta \in R$.
3. You should be able to judge how many solutions a linear system has, by using the row echelon form.

Typical Questions:

1.

Consider a linear system whose augmented matrix is of the form

$$\left[\begin{array}{ccc|c} 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 3 \\ 1 & 3 & a & b \end{array}\right]$$

(a) For what values of a and b will the system have infinitely many solutions?

(b) For what values of a and b will the system be inconsistent?

(a) $a = 5, b = 4$ (b) $a = 5, b \neq 4$

2.

Consider a linear system whose augmented matrix is of the form

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 2 & 5 & 3 & 0 \\ -1 & 1 & \beta & 0 \end{array}\right]$$

(a) Is it possible for the system to be inconsistent? Explain.

(b) For what values of β will the system have infinitely many solutions?

(a) Since the system is homogeneous it must be consistent.

1.3: Matrix Arithmetic

Vectors:

Vector is actually a special matrix with only one row or one column.

Example:

$$\vec{\mathbf{x}} = (x_1, x_2, x_3, x_4) \quad \text{and} \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ y_4 \end{bmatrix}$$

A m × n matrix A can be written as the set of vectors:

$$A = (\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n) \quad \text{or} \quad A = \begin{bmatrix} \vec{\mathbf{a}}_1 \\ \vec{\mathbf{a}}_2 \\ \vdots \\ \vec{\mathbf{a}}_m \end{bmatrix}, \text{ where } \vec{\mathbf{a}}_i = (a_{i1}, a_{i2}, \ldots, a_{in}) \qquad i = 1, \ldots, m$$

$$\mathbf{a}_j = \begin{bmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{bmatrix} \qquad j = 1, \ldots, n$$

Matrix representation of a linear system:

Consider now an $m \times n$ linear system

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned} \qquad \longrightarrow A\mathbf{x} = \mathbf{b}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

Linear Combination of Vectors (Important):

If $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$ are vectors in $\mathbb{R}^m$ and $c_1, c_2, \ldots, c_n$ are scalars, then a sum of the form

$$c_1\mathbf{a}_1 + c_2\mathbf{a}_2 + \cdots + c_n\mathbf{a}_n$$

is said to be a **linear combination** of the vectors $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$.

Consistency Theorem for Linear Systems

A linear system $A\mathbf{x} = \mathbf{b}$ is consistent if and only if $\mathbf{b}$ can be written as a linear combination of the column vectors of A.

Example:

$x_1 + 2x_2 = 5$

$x_1 - x_2 = -1$

$A = [\boldsymbol{a_1} \ \boldsymbol{a_2}] = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}$

It can be easily find that $1 \times \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2 \times \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 5 \\ -1 \end{bmatrix} \rightarrow 1 \times \boldsymbol{a_1} + 2 \times \boldsymbol{a_2} = \boldsymbol{b}$. Thus we can say that $\boldsymbol{b}$ can be written as the linear combination of $\boldsymbol{a_1}$ and $\boldsymbol{a_2}$, and (1,2) is one solution of the linear system.

Matrix Multiplication:

If $A = (a_{ij})$ is an $m \times n$ matrix and $B = (b_{ij})$ is an $n \times r$ matrix, then the product $AB = C = (c_{ij})$ is the $m \times r$ matrix whose entries are defined by

$$c_{ij} = \vec{\mathbf{a}}_i \mathbf{b}_j = \sum_{k=1}^{n} a_{ik} b_{kj}$$

Note: When you do matrix multiplication, check the **dimensions** of the two matrices first. The number of **columns** in A must be equal to the number of **rows** in B.

The matrix multiplication is not commutative, i.e., **AB≠BA**.

The Transpose of a Matrix:

The **transpose** of an $m \times n$ matrix A is the $n \times m$ matrix B defined by

$$b_{ji} = a_{ij}$$

for $j = 1, \ldots, n$ and $i = 1, \ldots, m$. The transpose of A is denoted by A^T.

Typical Questions:

1.

Let A be a 5×3 matrix. If

$$\mathbf{b} = \mathbf{a}_1 + \mathbf{a}_2 = \mathbf{a}_2 + \mathbf{a}_3$$

then what can you conclude about the number of solutions of the linear system $A\mathbf{x} = \mathbf{b}$? Explain.

Solution:

The given information implies that

$$\mathbf{x}_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{x}_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

are both solutions to the system. So the system is consistent and since there is more than one solution, the row echelon form of A must involve a free variable. A consistent system with a free variable has infinitely many solutions.

2.

Let

$$A = \begin{bmatrix} 1 & 2 \\ 1 & -2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}, \quad \mathbf{c} = \begin{bmatrix} -3 \\ -2 \end{bmatrix}$$

(a) Write $\mathbf{b}$ as a linear combination of the column vectors $\mathbf{a}_1$ and $\mathbf{a}_2$.

(b) Use the result from part (a) to determine a solution of the linear system $A\mathbf{x} = \mathbf{b}$. Does the system have any other solutions? Explain.

(c) Write $\mathbf{c}$ as a linear combination of the column vectors $\mathbf{a}_1$ and $\mathbf{a}_2$.

Solution:

(b) $\mathbf{x} = (2, 1)^T$ is a solution since $\mathbf{b} = 2\mathbf{a}_1 + \mathbf{a}_2$. There are no other solutions since the echelon form of A is strictly triangular.
(c) The solution to $A\mathbf{x} = \mathbf{c}$ is $\mathbf{x} = (-\frac{5}{2}, -\frac{1}{4})^T$. Therefore $\mathbf{c} = -\frac{5}{2}\mathbf{a}_1 - \frac{1}{4}\mathbf{a}_2$.

3.

Let $A\mathbf{x} = \mathbf{b}$ be a linear system whose augmented matrix $(A|\mathbf{b})$ has reduced row echelon form

$$\left[\begin{array}{ccccc|c} 1 & 2 & 0 & 3 & 1 & -2 \\ 0 & 0 & 1 & 2 & 4 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

(a) Find all solutions to the system.

(b) If

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ 3 \\ 4 \end{bmatrix} \quad \text{and} \quad \mathbf{a}_3 = \begin{bmatrix} 2 \\ -1 \\ 1 \\ 3 \end{bmatrix}$$

determine $\mathbf{b}$.

Solution:

(a) It follows from the reduced row echelon form that the free variables are x_2, x_4, x_5. If we set $x_2 = a$, $x_4 = b$, $x_5 = c$, then

$$\begin{aligned} x_1 &= -2 - 2a - 3b - c \\ x_3 &= 5 - 2b - 4c \end{aligned}$$

and hence the solution consists of all vectors of the form

$$\mathbf{x} = (-2 - 2a - 3b - c,\ a,\ 5 - 2b - 4c,\ b,\ c)^T$$

(b) If we set the free variables equal to 0, then $\mathbf{x}_0 = (-2, 0, 5, 0, 0)^T$ is a solution to $A\mathbf{x} = \mathbf{b}$ and hence

$$\mathbf{b} = A\mathbf{x}_0 = -2\mathbf{a}_1 + 5\mathbf{a}_3 = (8, -7, -1, 7)^T$$

1.4: Matrix Algebra

Algebraic Rules:

Each of the following statements is valid for any scalars α and β and for any matrices A, B, and C for which the indicated operations are defined.

1. $A + B = B + A$
2. $(A + B) + C = A + (B + C)$
3. $(AB)C = A(BC)$
4. $A(B + C) = AB + AC$
5. $(A + B)C = AC + BC$
6. $(\alpha\beta)A = \alpha(\beta A)$

7. $\alpha(AB) = (\alpha A)B = A(\alpha B)$
8. $(\alpha + \beta)A = \alpha A + \beta A$
9. $\alpha(A + B) = \alpha A + \alpha B$

The Identity Matrix:

The $n \times n$ **identity matrix** is the matrix $I = (\delta_{ij})$, where

$$\delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Note: 1. An identity matrix is a square matrix. 2. The elements on the diagonal are all 1s. 3. $I \times A = A, A \times I = A$.

Matrix Inversion:

An $n \times n$ matrix A is said to be **nonsingular** or **invertible** if there exists a matrix B such that $AB = BA = I$. The matrix B is said to be a **multiplicative inverse** of A.

Note: **nonsingular = invertible, singular = non-invertible**

A matrix can have at most one multiplicative inverse.

Algebraic Rules for Transposes and Inverse:

Algebraic Rules for Transposes

1. $(A^T)^T = A$
2. $(\alpha A)^T = \alpha A^T$
3. $(A + B)^T = A^T + B^T$
4. $(AB)^T = B^T A^T$

$$(A_1 A_2 \cdots A_k)^{-1} = A_k^{-1} \cdots A_2^{-1} A_1^{-1}$$

If $A^T = A$, we say A is a symmetric matrix.

Typical Questions:

1.

Let A be a nonsingular matrix. Show that A^{-1} is also nonsingular and $(A^{-1})^{-1} = A$.

Solution:

Since

$$A^{-1}A = AA^{-1} = I$$

it follows from the definition that A^{-1} is nonsingular and its inverse is A.

2.

Prove that if A is nonsingular, then A^T is nonsingular and

$$(A^T)^{-1} = (A^{-1})^T$$

[*Hint*: $(AB)^T = B^T A^T$.]

Solution:

Since

$$\begin{aligned} A^T(A^{-1})^T &= (A^{-1}A)^T = I \\ (A^{-1})^T A^T &= (AA^{-1})^T = I \end{aligned}$$

it follows that

$$(A^{-1})^T = (A^T)^{-1}$$

1.5: Elementary Matrices

Elementary Matrices:

An elementary matrix is obtained from the **identity matrix** by performing **exactly one elementary row operation**.

There are three types of elementary matrices corresponding to the three types of elementary row operations.

Type I An elementary matrix of type I is a matrix obtained by interchanging two rows of I.

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Type II An elementary matrix of type II is a matrix obtained by multiplying a row of I by a nonzero constant.

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Type III An elementary matrix of type III is a matrix obtained from I by adding a multiple of one row to another row.

$$E_3 = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Inverse of Elementary Matrices:

Theorem 1.5.1 If E is an elementary matrix, then E is nonsingular and E^{-1} is an elementary matrix of the same type.

Notes:

- If a matrix B is obtained by row operations E from A, i.e., B=EA, E^{-1} is actually the inverse operations to recover A from B.
- For Type I elementary matrices, $E^{-1}=E$
- For Type II elementary matrices, if the row operation from I to E is $4r_2$, then for E^{-1}, the corresponding operation is $1/4r_2$. (Just scaling it back)
- For Type II elementary matrices, if the row operation from I to E is $4r_2+r_1$, the corresponding operation is $-4r_2+r_1$. (Just change the sign)

Row Equivalent of Two Matrices:

A matrix B is **row equivalent** to a matrix A if there exists a finite sequence $E_1, E_2, \ldots, E_k$ of elementary matrices such that

$$B = E_k E_{k-1} \cdots E_1 A$$

I. If A is row equivalent to B, then B is row equivalent to A.

II. If A is row equivalent to B, and B is row equivalent to C, then A is row equivalent to C.

Equivalent Conditions for Nonsingularity (Important):

Let A be an $n \times n$ matrix. The following are equivalent:

(a) A is nonsingular.

(b) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\mathbf{0}$.

(c) A is row equivalent to I.

Corollary 1.5.3 The system $A\mathbf{x} = \mathbf{b}$ of n linear equations in n unknowns has a unique solution if and only if A is nonsingular.

Inverse Matrix Calculation (Important):

If A is nonsingular, then A is row equivalent to I and hence there exist elementary matrices $E_1, \ldots, E_k$ such that

$$E_k E_{k-1} \cdots E_1 A = I$$

Multiplying both sides of this equation on the right by A^{-1}, we obtain

$$E_k E_{k-1} \cdots E_1 I = A^{-1}$$

$(A|I) \rightarrow (I|A^{-1})$

Example:

Solve the system

$$\begin{aligned} x_1 + 4x_2 + 3x_3 &= 12 \\ -x_1 - 2x_2 &= -12 \\ 2x_1 + 2x_2 + 3x_3 &= 8 \end{aligned}$$

Solution

$$\left[\begin{array}{ccc|ccc} 1 & 4 & 3 & 1 & 0 & 0 \\ -1 & -2 & 0 & 0 & 1 & 0 \\ 2 & 2 & 3 & 0 & 0 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 3 & 1 & 0 & 0 \\ 0 & 2 & 3 & 1 & 1 & 0 \\ 0 & -6 & -3 & -2 & 0 & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 3 & 1 & 0 & 0 \\ 0 & 2 & 3 & 1 & 1 & 0 \\ 0 & 0 & 6 & 1 & 3 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 4 & 0 & \frac{1}{2} & -\frac{3}{2} & -\frac{1}{2} \\ 0 & 2 & 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 6 & 1 & 3 & 1 \end{array}\right]$$

$$\rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 2 & 0 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 0 & 6 & 1 & 3 & 1 \end{array}\right] \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ 0 & 1 & 0 & \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ 0 & 0 & 1 & \frac{1}{6} & \frac{1}{2} & \frac{1}{6} \end{array}\right]$$

Thus,

$$A^{-1} = \begin{bmatrix} -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{6} \end{bmatrix}$$

$$\mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -\frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \\ \frac{1}{4} & -\frac{1}{4} & -\frac{1}{4} \\ \frac{1}{6} & \frac{1}{2} & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 12 \\ -12 \\ 8 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ -\frac{8}{3} \end{bmatrix}$$

Diagonal and Triangular Matrices:

Upper triangular: $a_{ij} = 0$ for $i > j$

Lower triangular: $a_{ij} = 0$ for $i < j$

Typical Questions:

1.

Let

$$A = \begin{pmatrix} 2 & 1 \\ 6 & 4 \end{pmatrix}$$

(a) Express A as a product of elementary matrices.
(b) Express A^{-1} as a product of elementary matrices.

Solution:

A can be reduced to the identity matrix using three row operations

$$\begin{pmatrix} 2 & 1 \\ 6 & 4 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

The elementary matrices corresponding to the three row operations are

$$E_1 = \begin{pmatrix} 1 & 0 \\ -3 & 1 \end{pmatrix}, \quad E_2 = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}, \quad E_3 = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix}$$

So

$$E_3E_2E_1A = I$$

and hence

$$A = E_1^{-1}E_3^{-1}E_3^{-1} = \begin{pmatrix} 1 & 0 \\ 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$$

and $A^{-1} = E_3E_2E_1$.

2. Find the inverse of

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

Solution:

$$\begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix}$$

3.

Let A be a 3×3 matrix and suppose that

$$2\mathbf{a}_1 + \mathbf{a}_2 - 4\mathbf{a}_3 = \mathbf{0}$$

How many solutions will the system $A\mathbf{x} = \mathbf{0}$ have? Explain. Is A nonsingular? Explain.

Solution:

If we set $\mathbf{x} = (2, 1 - 4)^T$, then

$$A\mathbf{x} = 2\mathbf{a}_1 + 1\mathbf{a}_2 - 4\mathbf{a}_3 = \mathbf{0}$$

Thus $\mathbf{x}$ is a nonzero solution to the system $A\mathbf{x} = \mathbf{0}$. But if a homogeneous system has a nonzero solution, then it must have infinitely many solutions. In particular, if c is any scalar, then $c\mathbf{x}$ is also a solution to the system since

$$A(c\mathbf{x}) = cA\mathbf{x} = c\mathbf{0} = \mathbf{0}$$

Since $A\mathbf{x} = \mathbf{0}$ and $\mathbf{x} \neq \mathbf{0}$, it follows that the matrix A must be singular. (See Theorem 1.5.2)